

# Yu Li

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## Education

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| <b>George Washington University</b>                                                        | <b>Washington, D.C.</b>  |
| <i>Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0</i>                | <i>2025.08 – 2029.05</i> |
| Advisor: <a href="#">Prof. Tian Lan</a> 📧; Work with: <a href="#">Prof. Zhengling Qi</a> 📧 | <i>(expected)</i>        |
| <b>Wuhan University, Hongyi Honor College</b>                                              | <b>Wuhan, China</b>      |
| <i>B.Eng. in Microelectronics Science and Engineering, GPA: 3.87/4.0</i>                   | <i>2021.09 – 2025.06</i> |

## Industry Experience

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| <b>ByteDance Inc.</b>                                                                                                                                                                                                          | <b>San Jose, CA</b>      |
| <i>Machine Learning Scientist Intern, Tiktok Recommendation Core</i>                                                                                                                                                           | <i>2026.05 – 2026.08</i> |
| <ul style="list-style-type: none"> <li>Worked on production-scale video recommendation at TikTok.</li> <li>Built LLM-based ranking models for the ranking stage, integrating content signals into ranking features.</li> </ul> |                          |

## Research Highlights

- **LLM Reasoning:** Developing credit assignment, preference optimization, and context conditioning methods to improve alignment and reasoning in large language models [P1] [P3] [P4] [P5].
- **Agentic RL:** Designing reinforcement learning frameworks for hierarchical skill acquisition and multi-turn policy optimization across single and multi-agent systems [P2] [C2] [C5].
- **Efficient AI:** Improving training and inference efficiency for large language models through parameter-efficient fine-tuning and long-context KV cache management [C1] [C6] [C8].
- **Generative Models:** Exploring diffusion-based generative models for personalized content-style synthesis and cross-modal generation [C3] [J2].

## Preprints & Under Review

P=Preprint

- [P1] **OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning** 📧  
Yu Li, Rui Miao, Tian Lan, Zhengling Qi
- [P2] **ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning** 📧  
Yu Li, Rui Miao, Zhengling Qi, Tian Lan
- [P3] **Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO** 📧  
Yu Li, Tian Lan, Zhengling Qi
- [P4] **INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization** 📧  
Yu Li, Tian Lan, Zhengling Qi  
 AI with Recursive Self-Improvement Workshop at ICLR, 2026
- [P5] **Hindsight Self-Distillation: Demonstrating to the Teacher for On-Policy LLM Reasoning** 📧  
Yu Li, Shu Hong, Tian Lan  
*Under review at EMNLP 2026, OA: 3.33/5*

## Publications

C=Conference, J=Journal, †=Equal Contribution

I have published **6** papers in top-tier machine learning and computer vision venues (e.g., COLM, ACL, CVPR, AAAI, MM), with **4** of them as **first author**.

- [C1] **MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference**  
[COLM, 2026](#)   
Yu Li, Binxu Li, Tian Lan
- [C2] **Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization**  
[ACL Findings, 2026](#)   
Yu Li, Sizhe Tang, Tian Lan
- [C3] **CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion**  
[CVPR, 2026](#)   
Yu Li, Yujun Cai, Chi Zhang
- [C4] **KG-SAM: Encoding Structural Constraints into Segment Anything Models via Probabilistic Graphical Models**  
[ICASSP \(Oral\), 2026](#)   
Yu Li, Da Chang, Xi Xiao
- [C5] **ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense**  
[INFOCOM Intelligent Cloud Computing and Networking, 2026](#)   
Yu Li<sup>†</sup>, Sizhe Tang<sup>†</sup>, Rongqian Chen, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D. Bastian, Tian Lan
- [C6] **Mixed Text Recognition with Efficient Parameter Fine-Tuning and Transformer**  
[ICONIP, 2024](#)   
Yu Li<sup>†</sup>, Da Chang<sup>†</sup>
- [C7] **Prompted Information Bottlenecks: Rethinking Layer-Wise Information Allocation in Frozen Vision Transformers**  
[ACM MM, 2026](#)  
Yuqi Li, Xi Xiao, Yunbei Zhang, Lin Zhao, Yu Li, Aiden Zhao, Tianyang Wang, Hao Xu, Yingli Tian
- [C8] **Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT**  
[AAAI, 2026](#)   
Da Chang, Peng Xue, Yu Li, Yongxiang Liu, Pengxiang Xu, Shixun Zhang
- [J1] **Dual branch SAM-Transformer Fusion Network for Accurate Breast Ultrasound Image Segmentation**  
[Medical Physics, 2025](#)   
Yu Li, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*
- [J2] **SfMDiffusion: Self-supervised Monocular Depth Estimation in Endoscopy Based on Diffusion Models**  
[International Journal of Computer Assisted Radiology and Surgery, 2025](#)   
Yu Li, Da Chang, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*

## Selected Awards & Honors

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- **ICML 2026 Gold Reviewer**, top reviewers recognition. *2026*
- **Innova International Exchange Scholarship**, 6 recipients university-wide, ¥70,000. *2024*
- **Innova Excellence Scholarship**, Top 3%, ¥10,000, twice. *2023, 2024*
- **First-Class Scholarship**, Top 5%, ¥3,000, three times. *2022, 2023, 2024*

## Academic Services

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- **Conference Reviewer:** {NeurIPS, COLM, ACM MM, ICML, ICLR, AAAI, ICASSP}'2026
- **Journal Reviewer:** TMLR, TPAMI, ToN, Neurocomputing